BRACKET

		Max Stress					Max Deflection					Model Mass		
		Magnitude (psi)	X location	Y location	Z Location	Node #	Magnitude(inches)	X location	Y location	Z Location	Node #	lbs	Number of Elements	Factor of Safety
Inner Loop 1	Course	184773	0	-0.0561834	0.41705	2377	0.272673	0.8	0	-0.337	2512		1249	0.22
	Medium	197534	0	-0.03371	0.41705	13271	0.282056	0.8	0	-0.337	4		7998	0.2
	Diff	12761					0.009383						6749	0.02
	%Dif	6%					3%							9%
	Average % Diff	6%		Converge?	Yes									
	Fine (if needed)	222927	0	0	0.033835	80394	0.287009	0.8					75281	0.18
	Diff	25393					0.004953							0.02
	%Dif	11%					2%							9%
	Average % Diff	7%		Converge?	No									
	Design 1	197534	0	-0.03371	0.41705	13271	0.282056	0.8	0	-0.337	4	0.23728373	7998	0.2

Change Log

The FEA analysis convereged for Stress and Deflection but not exactly for safety factor. This is most likely because there was some miscalculation for the thickness in component "B" of the model causing very high stress concentrated at the junction of component "B" and "A". A finer mesh was created for acedemic rigor resulting in the stress perecent difference being higher between the fine and medium mesh than between the medium and coarse mesh. The deflection, however, was slightly closer. The safety factor was different by the same amount between both tests. The heat maps of stress and deflection were largely blue outside of the maximum locations, therefore the design change will focus there first. The remedy was decided to be mirroring feature B along the y axis. This eliminates the bending moment of the load and reduced stress greatly

Inner Loop 2	Course	8441.67	0	-0.3371	0.875	2577	0.001296	0.4	0.331979	0.05369	32		1464	4.7
	Medium	11314.9	0.8	-0.3371	0.9375	9233	0.0013652	0.4	0.335254	0.035237	12466		7520	3.5
	Diff	2873.23					6.915E-05	0	0.003275	0.018453				1.2
	%Dif	25%					5%							26%
	Fine (if needed)	15218.7	0	0.337	0.967	87218	0.0013979	0.4	-0.335254	0.035237	334		62871	2.6
	Diff	3903.8					0.0001019							0.5
	%Dif	26%					7%							11%
	Design 2	1521.87	0	0.337	0.967	87218	0.0013979	0.4	-0.335254	0.035237	334	0.27105857	62871	2.6

Change Log

Convergence was not achieved throughout FEA analysis for Design 2. It is suspected that stress concentrations at sharp corners of model are to blame. As the mesh gets finer and finer, the calculated stress gets hisgher and higher in the small corners leading to ever ascending readings for stress and safety factor. The design change was very effective reducing the maximum stress by a factor of 129 (2 orders of magnitude), and the maximum deflection by a factor of 117.6. The next design change will be to fillet the corners of the part in an attempt to get the FEA to converger itself like a good little boy. Additionally through holes will be added to lighten the part and the thickness of component "C" will be reduced to the same end.

Inner Loop 3	Course	8313.56	-0.14734	0.22365	-0.039436	11	0.0008435	0.4	0	-0.2271	129		2323	4.8
	Medium	10123.4	0.156	-0.2271	0	156	0.0008729	0.4	0	-0.2271	15524		9248	3.9
	Diff	1809.84					2.943E-05							0.9
	%Dif	18%					3%							19%
	Fine (if needed)	10621.3	-0.14734	-0.225157	-0.029643	88182	0.0008845	0.383902	0	-2.51763	87883		55976	3.8
	Diff	497.9					1.156E-05							0.1
	%Dif	5%					0%							3%
	Design 3	10621.3	-0.14734	-0.225157	-0.029643	88182	0.0008845	0.383902	0	-2.51763	87883	0.22337255	55976	3.8

Change Log

LINK

		Max Stress					Max Deflection					Model Mass		
		Magnitude (psi)	X location	Y location	Z Location	Node #	Magnitude(inches)	X location	Y location	Z Location	Node #	lbs	Number of Elements	Factor of Safety
Inner Loop 1	Course	25497.5	0.025	-0.499154	-0.558677	3899	0.0021067	0	0	-1.08775	3766		1817	4.7
	Medium	25789	0	-0.5	-0.58775	15980	0.0021087	0.05	0	-1.08775	47		8041	4.7
	Diff	291.5					1.93E-06							0
	%Dif	1%					0%							0%
	Average % Diff	0%		Converge?	Yes									
	Fine (if needed)													
	Diff	25789					0.0021087							4.7
	%Dif													
	Average % Diff			Converge?	No									
	Design 1	25789	0	-0.5	-0.58775	15980	0.0021087	0.05	0	-1.08775	47	0.03	8041	4.7

Change Log

For the most part the FEA reflected previous calculations and the Factor of safety was upheld at or above 4. There is, however about 0.002" of deflection. In order to remediate The thickness will be increased from 0.050 to 2mm (0.078"). Also, because the "B" component was doubled the link can no longer slide into place. There is now a hole through "A" that can accomodate a pin with a diameter of 0.4542". This is still in the same fit classification and schedule of the previous hole so the diameter of of the bracket side of the link will be decreased to 0.45801.

2	Course	19045.5	0.025	-0.497261	-0.425136	1116	0.001439	0.05	0	-0.977401	1060		538	4.6
	Medium	19816.5	0.05	0.05	-0.477401	14235	0.001496	0.05	0	-0.977401	103		8130	4.5

Inner Loop	Diff	771					0.000057	0	0	0				0.1
	%Dif	4%					4%							2%
	Fine (if needed)													
	Diff	19816.5					0.001439							0.5
	%Dif													11%
	Design 2	26816.5	0.05	0.05	-0.477401	14235	0.0023173	0.05	0	-0.977401	103	0.05	8130	4.5

Change Log The deflection is still do damn high.... Like the rent... and my uncle steve. Thickening to 2.5mm (0.098")

Inner Loop 3	Course	13452.4	0	-0.495722	-0.41237	2620	0.0011755	0	0	-0.977401	55		1299	8.9
	Medium	13539.7	0.04921	0.499695	-0.49485	13684	0.0011783	0	0	-0.977	14848		8889	8.9
	Diff	87.3					2.77E-06							0
	%Dif	1%					0%							0%
	Fine (if needed)													
	Diff	13539.7					0.0011783							8.9
	%Dif													
	Design 3	13539.7	0.04921	0.499695	-0.49485	13684	0.0011783	0	0	-0.977	14848	0.06	8889	8.9